Sketching Self-Guide

Metals, Monday 11/2. Use this if you get stuck on the sketch packet. It is the slides from class, with notes.

Today, in order

1. Journal and phones in the caddy. No shop floor, no tools.
2. Read pages 2 to 6 of this guide, or follow the slides on the screen.
3. Sketch packet: Examples A to C first. They are already drawn, so study how they work.
4. Problems 1 to 4. Metals 2 and 3, and anyone who finishes, go on to Problems 5 to 7.
5. Tick the Me box in each CHECK IT box. Leave the Teacher box blank.
6. Packet goes in the period folder, finished or not.
7. Tonight, by 11:59: photo your packet pages and upload them to **Sketch Sheet 1: Iso and Ortho** in Canvas.

Two short videos

Scan with your phone or click the links in the Canvas module **Sketching Is a Language**. Headphones, or watch with captions.

Orthographic Drawing, Simplified (2 min, Paxton Patterson)
Top, front and right views, and how they line up. Watch this before Problem 1.
`https://www.youtube.com/watch?v=SdLegfoMXNA`

Beginner Isometric Drawing Tutorial (5 min, Ms. Cooper's Art Class)
How to draw the picture on a triangle grid. Watch this before Problem 5.
`https://www.youtube.com/watch?v=biz81MlBFZo`

Who to ask

1. This guide and the videos.
2. A Metals 2 or 3 student at your table.
3. The adult in the room. Mr. Kirsch is next door in F112.

Name this object

A cylindrical vessel of glazed ceramic, 85 mm in outside diameter and 100 mm tall, with a wall 6 mm thick and a flat, closed base. The top is open. A single C-shaped loop of elliptical section, 10 mm by 14 mm, is joined to the outer wall at two points, 25 mm and 75 mm below the rim, and projects 35 mm from the wall. It holds 350 millilitres.

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SLIDE 2

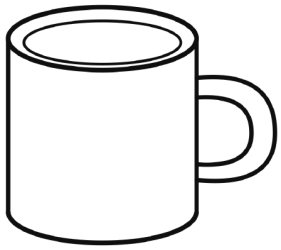
Read this before you look at the next slide. Try to name the object. Every number is true and it still does not help much.

It's a coffee mug.

70 words, and you were still guessing.

One sketch. Nobody asks.

Words are slow and everyone pictures something different. A sketch is one look, and we all see the same thing.



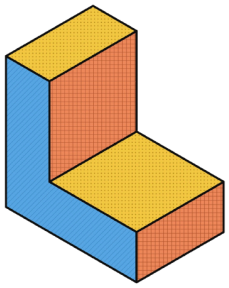
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SLIDE 3

A coffee mug. Seventy words and most people are still guessing. One quick sketch and nobody has to ask. That is why we sketch.

Sketching is a language

1. **Words**
   Slow. Everyone pictures something different.
2. **A sketch**
   One look. Same picture for everyone in the shop.
3. **Grammar**
   A language has rules: views, line types, grids. That is what we learn here.



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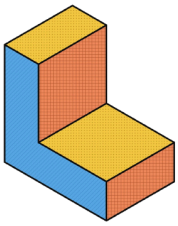
SLIDE 4

Sketching is a language, and a language has rules. The rules are views, line types and grids. Follow them and a welder in another shop can build your part from your sketch.

Two ways to say the same block

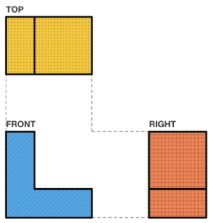
ISOMETRIC

The picture. Three faces at once. Drawn on the triangle grid.



ORTHOGRAPHIC

One face at a time, straight on. Drawn on the square grid.

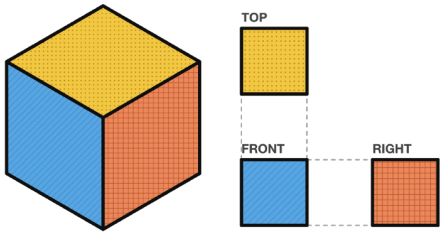


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SLIDE 5

Same block, two ways. Isometric is the picture: you see three faces at once, on the triangle grid. Orthographic is one face at a time, straight on, on the square grid.

Orthographic means one face at a time



Look straight at one face. You see a flat shape with no depth.

Top, front, right: three looks, three flat shapes. The colour tells you which face you are looking at.

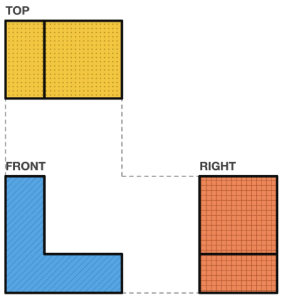
TOP
FRONT
RIGHT

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SLIDE 6

Look straight at one face. You see a flat shape with no depth. Do that from the top, the front and the right and you get three flat shapes. On the sheet, top is dots, front is diagonal lines, right is cross-hatch.

The three views line up



TOP sits straight above FRONT. They share width.

RIGHT sits straight beside FRONT. They share height.

TOP and RIGHT share depth.

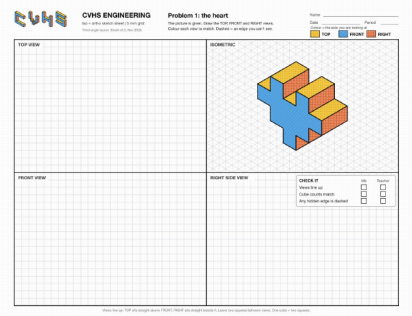
Count cubes. Do not guess.

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SLIDE 7

The views line up. TOP goes straight above FRONT, so they are the same width. RIGHT goes straight beside FRONT, so they are the same height. Count cubes. Do not guess.

Your sheet



Square grid, three views. TOP above FRONT, RIGHT beside it.

Triangle grid, top right. The isometric picture goes here.

One cube = two squares. Two triangle edges on the iso grid. Same size everywhere, so nothing needs measuring.

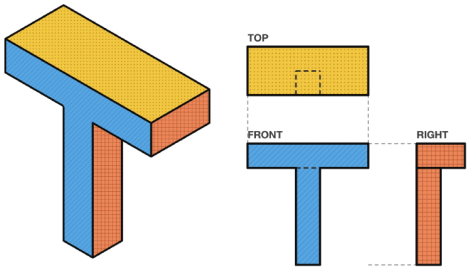
Check it box. Tick yours before you hand it in.

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SLIDE 8

Your sheet. The square grid holds the three views. The triangle grid, top right, holds the picture. One cube is two squares. Tick the Me box in CHECK IT before you turn it in.

Dashed means hidden



The stem is tucked under the bar. From the top you cannot see it, but it is still there.

An edge you cannot see is drawn dashed.

Solid line: an edge you can see. Dashed line: an edge behind a face.

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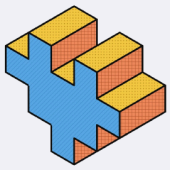
SLIDE 9

Dashed means hidden. If an edge is behind a face and you cannot see it, draw it dashed. Solid lines are edges you can see. Example C shows this.

Both directions, today

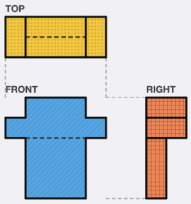
Picture given. Draw the three views.

Problems 1 to 4: heart, 6, 7, ghost



Views given. Draw the picture.

Problems 5 to 7: mushroom, trophy, cactus



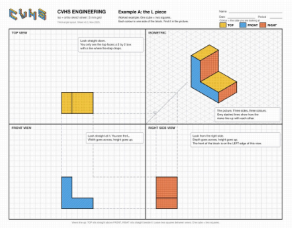
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SLIDE 10

Problems 1 to 4 give you the picture and you draw the three views. Problems 5 to 7 give you the views and you draw the picture.

How to sketch on the grid

1. **Pencil, light lines first.** You will fix things. Make that cheap.
2. **Count cubes from what is given.** Two squares each. Width, height, depth.
3. **Darken the lines you are sure of.** Corners sharp, lines on the grid.
4. **Dashed for anything hidden.** A face in front of an edge: dash the edge.
5. **Line the views up before you call it done.** TOP over FRONT, RIGHT beside it.



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SLIDE 11

Light pencil first, so mistakes are cheap. Count cubes. Darken what you are sure of. Dash what is hidden. Check that the views line up.

Stuck? Find your problem

I do not know where to start.	Go to Example A. It is already drawn. Count the cubes in the picture, then count the squares in the front view. One cube is two squares.
My views do not line up.	TOP must sit straight above FRONT. Put your pencil on the left edge of FRONT and slide it straight up. It should hit the left edge of TOP.
My shape is the wrong size.	You guessed. Count cubes across, up and back, then double it for squares.
I cannot tell top from front.	Front is the face you would look at if you stood in front of the part. Top is what a bird sees. Right is what you see walking around to the right.
A line is behind something.	Draw it dashed. Short dashes, small gaps.
Still stuck.	Watch the video for that problem, then ask the adult in the room or a Metals 2 or 3 student. Then try again.

Before you turn it in: TOP above FRONT, RIGHT beside FRONT, cubes counted, hidden edges dashed, Me box ticked.

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